

Companion XLIV: Gravitational Slip in the Relaxation-Driven Cyclic Cosmology

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Abstract

The gravitational slip, defined as the difference between the two scalar metric potentials in the Newtonian gauge, provides a sensitive probe of the dynamical structure of spacetime. In the standard cosmological model, the slip is negligible in the absence of anisotropic stress, and the two potentials evolve identically. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential introduces a natural gravitational slip, as the relaxation mechanism modifies the temporal evolution of the potentials in a scale-dependent manner. This Companion develops a continuous description of the gravitational slip in RDCC, deriving how the relaxation scale influences the metric potentials, the anisotropic stress, and the observational signatures in weak lensing, redshift-space distortions, and the integrated Sachs–Wolfe effect. The resulting framework provides a new geometric perspective on the RDCC Universe.

1 Introduction

In the Newtonian gauge, scalar perturbations of the metric are described by two potentials, Φ and Ψ , which govern the motion of matter and the propagation of light. In the standard cosmological model, these potentials are equal in the absence of anisotropic stress, and their evolution is governed by the linear growth of structure.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the temporal evolution of the metric potentials in a scale-dependent manner, producing a natural gravitational slip. The slip becomes a direct tracer of the relaxation scale k_{rel} and the temporal structure of the RDCC gravitational field.

2 Metric Potentials in RDCC

The perturbed metric in the Newtonian gauge is

$$ds^2 = -(1 + 2\Psi) dt^2 + a^2(t)(1 - 2\Phi) d\mathbf{x}^2. \quad (1)$$

In RDCC, the relaxation mechanism modifies the evolution of the potentials. The potential Φ evolves according to

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right], \quad (2)$$

while the potential Ψ evolves as

$$\Psi_{\text{RDCC}}(k, a) = \Psi(k, a) \left[1 - \chi_{\Psi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

Because the relaxation mechanism affects the two potentials differently, a gravitational slip arises naturally.

3 Definition and Evolution of the Gravitational Slip

The gravitational slip is defined as

$$\eta(k, a) = \frac{\Phi_{\text{RDCC}}(k, a)}{\Psi_{\text{RDCC}}(k, a)}. \quad (4)$$

In the standard cosmological model, $\eta = 1$ in the absence of anisotropic stress. In RDCC, the slip becomes

$$\eta_{\text{RDCC}}(k, a) = \frac{1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha}}{1 - \chi_{\Psi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha}}. \quad (5)$$

This expression shows that the slip is scale-dependent and becomes most pronounced near the relaxation scale. The slip reflects the temporal structure of the RDCC gravitational field and provides a direct geometric signature of the relaxation mechanism.

4 Anisotropic Stress and the RDCC Geometry

In general relativity, a gravitational slip can arise from anisotropic stress. In RDCC, the slip arises even in the absence of physical anisotropic stress, because the relaxation mechanism modifies the temporal evolution of the metric potentials in a scale-dependent manner.

The effective anisotropic stress can be defined as

$$\pi_{\text{eff}}(k, a) = \frac{k^2}{4\pi G a^2} [\Phi_{\text{RDCC}}(k, a) - \Psi_{\text{RDCC}}(k, a)]. \quad (6)$$

This effective stress encodes the geometric imprint of the relaxation mechanism and provides a new perspective on the structure of spacetime in RDCC.

5 Observational Signatures

The gravitational slip affects several key observables. Weak lensing is sensitive to the combination $\Phi + \Psi$, while galaxy motions are sensitive to Ψ alone. The slip therefore modifies the relation between lensing and clustering.

The integrated Sachs–Wolfe effect is sensitive to the time derivative of the metric potentials, and the slip introduces a scale-dependent modification to the ISW signal.

Redshift-space distortions depend on the velocity field, which is governed by Ψ . The slip therefore introduces a scale-dependent modification to the RSD signal.

These signatures provide a direct observational window into the relaxation scale and the geometric structure of the RDCC Universe.

6 Conclusion

The gravitational slip in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the metric potentials. The relaxation mechanism modifies the temporal evolution of Φ and Ψ in a scale-dependent manner, producing a natural gravitational slip even in the absence of anisotropic stress. This slip affects weak lensing, redshift-space distortions, and the integrated Sachs–Wolfe effect, providing a direct geometric signature of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of the geometry of spacetime, the interplay between matter and light, and the observational tests of RDCC.

References

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